0.1 (1)

The angular momentum of the small body is

$$\vec{J} = \vec{r} \times \vec{p} = m\vec{r} \times \dot{\vec{r}}.$$

Taking its time derivative, we obtain

$$\begin{aligned} \frac{d\vec{J}}{dt} &= m\dot{\vec{r}} \times \dot{\vec{r}} + m\vec{r} \times \ddot{\vec{r}} \\ &= \vec{r} \times \vec{F}(\vec{r}). \end{aligned}$$

Since the gravitational force is a central force, $\vec{F}(\vec{r})$ is parallel to $\vec{r}$, so

$$\frac{d\vec{J}}{dt} = \vec{r} \times \vec{F}(\vec{r}) = 0.$$

Thus, $\vec{J}$ is conserved. Since $\vec{J}$ is always perpendicular to $\vec{r}$, the motion is confined to a fixed plane passing through the Sun.

0.2 (2)

The Lagrangian in polar coordinates is

$$L = K - V = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{GMm}{r}.$$

Applying the Euler–Lagrange equation to θ,

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} &= 0, \\ \frac{d}{dt}\left(mr^2\dot{\theta}\right) &= 0. \end{aligned}$$

Using the definition $h = r^2\dot{\theta}$, we obtain

$$\frac{d}{dt}(mh) = 0 \quad \Longrightarrow \quad \frac{dh}{dt} = 0.$$

Thus, h is conserved.

0.3 (3)

Using $\dot{\theta} = h/r^2$, the total energy becomes

$$\begin{aligned} E &= K + V \\ &= \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - \frac{GMm}{r} \\ &= \frac{1}{2}m\left(\dot{r}^2 + \frac{h^2}{r^2}\right) - \frac{GMm}{r}. \end{aligned}$$

0.4 (4)

Notice that the energy is now expressed entirely in terms of r and $\dot{r}$. Introducing the effective potential

$$V_{\text{eff}}(r) = \frac{mh^2}{2r^2} - \frac{GMm}{r},$$

we can rewrite the energy as

$$E = \frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r).$$

Since the radial kinetic energy is always non-negative,

$$E - V_{\text{eff}}(r) = \frac{1}{2}m\dot{r}^2 \geq 0, \qquad E \geq V_{\text{eff}}(r).$$

Therefore, the minimum allowed energy is the minimum of $V_{\text{eff}}(r)$. Since V_{eff} is a quadratic function of $1/r$, we can find its minimum by solving

$$\begin{aligned} \left.\frac{dV_{\text{eff}}}{dr}\right|_{r=r_0} &= 0, \\ -\frac{mh^2}{r_0^3} + \frac{GMm}{r_0^2} &= 0, \\ r_0 &= \frac{h^2}{GM}. \end{aligned}$$

Substituting this into the effective potential gives

$$E_0 = V_{\text{eff}}(r_0) = \frac{mh^2}{2r_0^2} - \frac{GMm}{r_0} = \boxed{-\frac{G^2M^2m}{2h^2}}.$$

0.5 (5)

At the maximum and minimum distances, r_1 and r_2, the radial velocity vanishes, $\dot{r} = 0$. Therefore,

$$\begin{aligned} E &= V_{\text{eff}}(r), \\ E &= \frac{mh^2}{2r^2} - \frac{GMm}{r}, \\ 2Er^2 + 2GMmr - mh^2 &= 0. \end{aligned}$$

Solving this quadratic equation for r, we obtain

$$\begin{aligned} r_{1,2} &= -\frac{GMm}{2E}\left(1 \pm \sqrt{1 + \frac{2Eh^2}{G^2M^2m}}\right) \\ &= \boxed{-\frac{GMm}{2E}\left(1 \pm \sqrt{1 - \frac{E}{E_0}}\right)}. \end{aligned}$$

Here, the plus sign gives the maximum distance r_1, while the minus sign gives the minimum distance r_2.

0.6 (6)

The time-averaged kinetic energy is

$$\begin{aligned}\langle K\rangle &= \left\langle \frac{1}{2}m\dot{\vec{r}}\cdot\dot{\vec{r}}\right\rangle \\ &= \frac{1}{2}\left\langle \frac{d}{dt}\left(m\vec{r}\cdot\dot{\vec{r}}\right) - m\vec{r}\cdot\ddot{\vec{r}}\right\rangle \\ &= \frac{1}{2}\left\langle \frac{d}{dt}\left(\vec{r}\cdot\vec{p}\right) - \vec{r}\cdot\vec{F}(\vec{r})\right\rangle .\end{aligned}$$

For a bound orbit ($E<0$), $\vec{r}$ and $\vec{p}$ remain bounded, so the time average of the total derivative vanishes:

$$\left\langle \frac{d}{dt}(\vec{r}\cdot\vec{p})\right\rangle = \lim_{T\to\infty}\frac{[\vec{r}\cdot\vec{p}]_0^T}{T} = 0.$$

Hence,

$$\boxed{\langle K(\vec{r})\rangle = -\frac{1}{2}\langle\vec{r}\cdot\vec{F}(\vec{r})\rangle}, \qquad \boxed{A=-\frac{1}{2}}.$$

0.7 (7)

For the gravitational force,

$$\vec{F}(\vec{r}) = -\frac{GMm}{r^3}\vec{r},$$

so

$$\vec{r}\cdot\vec{F}(\vec{r}) = -\frac{GMm}{r} = V(r).$$

Thus, from (6),

$$\langle V\rangle = -2\langle K\rangle.$$

Since the total energy is conserved,

$$E = \langle K\rangle + \langle V\rangle = \langle K\rangle - 2\langle K\rangle = \boxed{-\langle K\rangle}, \qquad \boxed{C=-1}.$$

0.8 (8)

For $E=0$, the quadratic equation in (5) reduces to

$$2GMmr - mh^2 = 0,$$

which has one positive turning point. The corresponding Kepler orbit is a parabola.

For $E>0$, the same quadratic equation has one positive root and one negative root. Only the positive root is physically relevant, and the corresponding orbit is a hyperbola.

In both cases, the small body can escape to infinity. As $r\to\infty$, the gravitational potential energy tends to zero, so

$$K = E - V(r) \longrightarrow E.$$

Therefore, the long-time average of the kinetic energy is

$$\boxed{E = \langle K\rangle \qquad (E\geq 0)}.$$

In particular, $\langle K\rangle = 0$ for a parabolic orbit ($E=0$).